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Improving failure modeling for gas transmission pipelines: A survival analysis and machine learning integrated approach

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A R T I C L E I N F O

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A B S T R A C T

This study proposes a methodology to model gas transmission pipeline failures using historical pipeline failure data. Censoring occurs frequently in the dataset, and overlooking it may lead to biased predictions. To address this issue, the statistical Cox model, a survival analysis and machine learning integrated model, i.e., RSF, are introduced in this study, along with other machine learning models (ANN, SVR, RF, and XGBoost), primarily for comparison. The Cox and RSF models provide insights into the influence of covariates on pipeline failure, informing decisions regarding pipeline construction, inspection, and maintenance activities. The findings indicate that the statistical Cox model overestimates failure age due to its limited ability to capture failure nonlinearity, while other machine learning models underestimate failure age because they cannot handle dataset censoring. In contrast, the survival analysis integrated machine learning method, RSF, outperforms other methods for modeling gas pipeline failures. The findings have practical implications for effectively managing reliability and mitigating risks associated with gas transmission pipelines to ensure safety. Moreover, the proposed methodology can potentially be applied to other pipeline systems and various types of systems, provided certain requirements are met.

1. Introduction

Natural gas is widely considered as a cleaner fuel compared to coal and petrol due to its less carbon dioxide emissions. It plays a crucial role in meeting the energy demands of modern cities for both production and living activities. In the U.S., natural gas pipelines are seen as a significant contributor to achieving the ambitious climate goal of reducing greenhouse gas emissions by more than 50% by 2030. Pipeline infrastructure is an integral part of the oil and gas industry, facilitating the transportation of gas from production sites to refineries, processing plants, and end-users in a safe, continuous, and reliable manner. Currently, the U.S. has approximately 320,000 miles of natural gas transmission pipelines in operation. However, there have been over 5500 incidents involving pipelines, resulting in nearly 600 injuries and 125 fatalities in recent decades [1,2]. Therefore, ensuring the safety of pipelines is of utmost importance to relevant authorities, and developing effective and reliable prediction models is a critical research area.

Numerous models have been proposed to date to model the deterioration process of pipelines and assess the associated risk of failure. These models can generally be classified into three categories: physical, statistical, and machine learning models [3]. Statistical models are built using historical data on pipeline failures to identify patterns and represent them mathematically. This allows for a proper assessment of the contribution of pipe characteristics, operational and environmental factors, as well as the prediction of future failures [4]. For instance, Caleyo et al. (2008) estimated the annual failure rate of oil and gas pipeline systems using the homogeneous Poisson process (HPP) and the power law process. This enabled the depiction of the failure process with both constant and varying intensity [5]. Pesis and Tee (2017) employed a parametric hybrid empirical hazard model and nonlinear quantile regression to examine pipeline failure records from 2002 to 2014. Their findings provided valuable insights into the performance of existing pipelines [6]. Many studies have attempted to evaluate the failure probability and risk of pipelines based on historical failure data. In these studies [7–10], modification factors were introduced to adjust the baseline probability associated with each failure cause, taking into account the characteristics of the target pipeline. However, this method falls short in capturing the time-varying nature of pipelines and their

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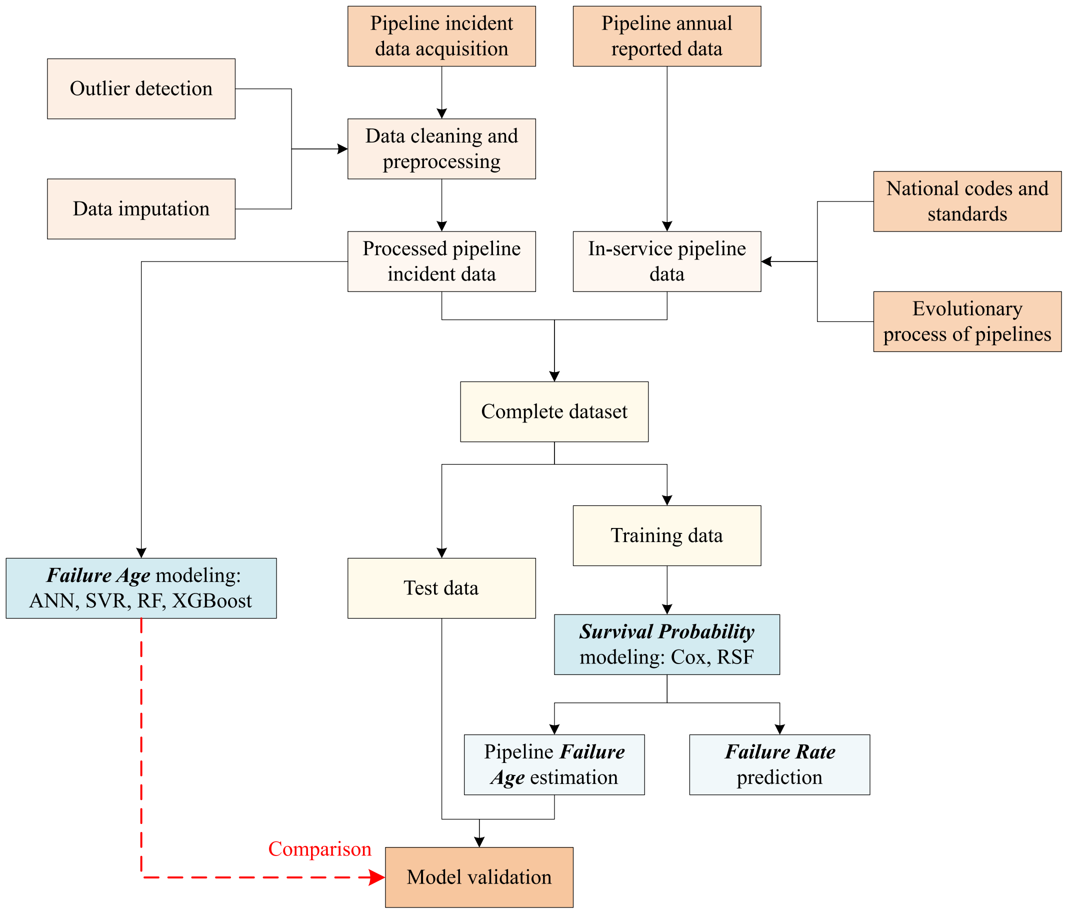


Fig. 1. Flowchart of the proposed methodology.

related attributes. Statistical models also struggle to fully consider and reflect the complex nonlinear relationship between various parameters and the outcomes present in the pipeline failure process. Nevertheless, in specific scenarios, statistical models remain a viable option, provided there is an adequate amount of reliable failure records.

Physical models attempt to investigate failure mechanisms and understand the capacity of pipelines to resist failure [4]. To evaluate failure probability or reliability, a limit state function is based on pipe wall thickness, stress, or internal pressure commonly employed [11,12]. Various computational approaches are used to estimate reliability, with the Monte Carlo simulation (MCS) being the most commonly utilized method [13]. Other techniques, such as importance sampling (IS) and subset simulation (SS), have also been developed to enhance efficiency and accuracy, especially in cases with very low failure probabilities [14–16]. The first-order reliability method (FORM) is an alternative to MCS and has been explored for calculating the reliability of corroded gas pipelines [12]. Structural reliability analysis is crucial for accurate reliability assessments in gas pipelines, particularly when comprehensive data on corrosion defects, including details like depth, width, corrosion rate, and historical inspection records, are available. While this analysis is typically conducted at the individual pipe level, applying it to a network poses challenges due to the large number of pipelines and the computational time required, especially when only partial information is accessible. Moreover, the complex deterioration and failure mechanisms of pipelines lack a comprehensive understanding.

Machine learning methods have garnered significant attention in research fields due to their ability to analyze complex relationships between inputs and targets. These methods also excel in handling uncertainty, fuzziness, and ambiguity. Consequently, they have been extensively applied in pipeline failure modeling. For example, El-Abbasy et al. (2014) developed models using artificial neural networks (ANNs) based on historical inspection data to evaluate and predict the condition of oil and gas pipelines. These models achieved an average accuracy of over 97% in the validation dataset [17]. Another study by Yang et al. proposed a data-driven model that incorporated graph embedding and a deep clustering algorithm to assess the consequences of an urban gas pipeline network. The effectiveness of this model was verified through validation on a real gas pipeline network with more than 6500 pipelines [18]. In addition, Phan et al. utilized ANNs as surrogate models to calculate the peak tensile and compressive strains of steel pipelines [19]. Similar to statistical models, machine learning models rely on failure data to identify patterns. The quantity and quality of available data play a crucial role in the accuracy and applicability of these models. Historical failure records collected and maintained by relevant government departments or companies are commonly used due to their accessibility.

In statistical and machine learning research, modeling the failure probability or reliability using historical data often involves encountering censored data. Censored data occurs when only a subset of pipelines have experienced failures, while the majority remain in service until the observation period ends. Failure to account for censoring can lead to biased predictions [20]. Survival analysis is a statistical model designed to handle time-to-failure data and has been utilized in water pipe failure modeling [21–25]. However, there has been limited consideration given to censoring when modeling gas pipeline failures based on historical data. Additionally, the statistical methods currently employed have limited ability to capture the nonlinearities inherent in the failure process of gas pipelines in relation to different covariates, resulting in subpar predictive performance. Furthermore, assuming a constant failure probability or failure rate throughout a pipeline's life cycle may not be appropriate [7–10]. Instead, the bathtub curve, which depicts a higher failure rate in the early stages of service, should be taken into account. This study aims to develop a reliable and efficient methodology for assessing the reliability of gas transmission pipelines by addressing censoring in the dataset and incorporating the dynamic failure rate across different ages to enhance failure modeling.

This study presents a methodology for modeling the failure probability of pipelines within a gas transmission network. Unlike structural reliability analysis methods based on physical failure mechanisms and probability theory, the approach utilizes historical failure data to estimate the failure probability for pipelines. The estimation of failure probability was improved by removing outliers and filling missing data using the MissForest method. As information regarding pipelines still in service, until the latest observation is confidential and inaccessible to the public, this study generated data concerning these pipelines from annual reports. Both a statistical model, the Cox proportional hazard model, and a machine learning model integrated with survival analysis, i.e., random survival forests (RSF), were developed using the complete dataset to estimate the failure probability of pipelines, while appropriately handling censoring information and introducing a dynamic failure rate with respect to pipeline ages. The Cox and RSF models provided ways to assess the impacts of various covariates on the failure probability. Additionally, various commonly used machine learning models, including ANN, Support Vector Regression (SVR), Random Forest (RF), and XGBoost, were employed for the modeling of gas pipeline failure time, using failure records solely. The performance of models was compared using the test data. The findings of this study demonstrate an efficient and reliable approach for assessing the reliability of gas pipelines using historical data. Furthermore, the proposed methodology holds potential applicability in other engineering systems and can inform corresponding practices for reliability and risk management.

2. Methodology

Fig. 1 presents the proposed methodology used in this study to model the failure probability of gas transmission pipelines. The details of each step are described in the following sections.

2.1. Data acquisition, processing and preparation

This study used failure records maintained by the Pipeline and Hazardous Materials Safety Administration (PHMSA) to develop models for the failure probability of gas transmission pipelines. Extensive incident-specific data were gathered in accordance with 49 CFR Part 191 by the local operator, encompassing details like pipe attributes, consequences (comprising costs, injuries, and fatalities), incident location, and more. Contributing factors were chosen based on domain expertise. Subsequently, a meticulous examination of the dataset was conducted to pinpoint outliers and missing data. The presence of outliers can be attributed to human error during manual recording. For instance, some records indicate implausible pipe thicknesses, such as 375 inches (likely intended to be 37.5 inches) or wall thicknesses exceeding the pipe diameter. Similarly, exceptionally high maximum yield strength values, surpassing normal levels, were identified as clearly implausible. To address these issues, erroneous entries were either adjusted to fall within the valid range or removed from the dataset altogether. It was discovered that certain missing values were also present. While some studies have advocated outright deletion of records with missing data [26,27], this approach not only sacrifices potentially valuable information but also leads to an underestimation of the gas pipeline hazard rate.

This study employed MissForest for missing data imputation due to its key advantages, including robustness in handling high-dimensional datasets, noise resilience, and the absence of strict assumptions. The MissForest algorithm proceeds as follows: initially, it imputes missing values by employing the mean (for continuous variables) and the most frequent class (for categorical variables); subsequently, it partitions the dataset into observed and missing data subsets, using the former for training and the latter for prediction. Both sets are then fed into a RF model, trained to predict the target variable using available data. The RF model generates predictions for missing values in the prediction set, replacing them and yielding a transformed dataset. This imputation process iterates for each variable with missing data, culminating in one imputation iteration. The iterations continue for multiple rounds, refining imputed values during each cycle. The process concludes when the relative sum of squared differences (for continuous variables) or the proportion of misclassified entries (for categorical variables) between the current and previous imputations increases, designating the previous imputation as final. MissForest thus iteratively enhances imputed values using the RF model, effectively leveraging available data to create a comprehensive dataset for further analysis, demonstrating its superior performance relative to widely used imputation methods [28]. Following the imputation of missing data, the failure records were deemed ready for subsequent model development.

Due to the absence of publicly available information on gas transmission pipelines currently in service, a simulation approach was employed in this study, utilizing annual reports submitted by operators to PHMSA. These reports offered the breakdown of miles of gas transmission on various pipeline attributes, including diameter, material, installation decade, and location class (detailed in Section 3), enabling the estimation of pipeline installations and their corresponding characteristics over time. The simulated dataset, in conjunction with failure records, was subsequently aggregated to construct the gas pipeline failure model.

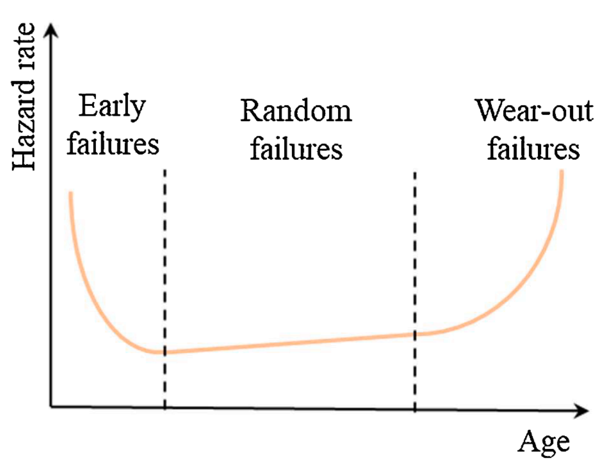


Fig. 2. Typical bathtub curve in the life cycle of products.

2.2. Background of survival analysis and machine learning

2.2.1. Survival analysis

Survival analysis typically focuses on time-to-event data, where data are generally not fully observed but censored. For a given pipe with a failure time T_i, and a non-negative censoring variable U that signifies the end of the study or the duration of follow-up, a censored failure time random variable $X_i = \min(T_i, U_i)$ is generally observed. The most common form of censoring is right censoring, as the failure of a pipeline has not happened at the end of the study, which is also the case considered in this study. The failure indicator δ_i is used to represent the status of a single pipeline [20]

$$\delta_i = \begin{cases} 1 \ if \ T_i \leq U_i \\ 0 \ if \ T_i > U_i \end{cases} \tag{1}$$

Then, the survival probability $S(t)$ and the failure probability $F(t)$ of a single pipe are defined as

$$S(t) = P(T > t) = \int_t^\infty f(u)\mathrm{d}u \tag{2}$$

$$F(t) = P(T \leq t) = 1 - S(t) \tag{3}$$

where $f(t)$ is the density function given by

$$f(t) = \lim_{\Delta t \to 0} \frac{1}{\Delta t} P(T \leq t < t + \Delta t) \tag{4}$$

It is also interesting to investigate the probability of failure occurring at time t when the pipeline has not failed yet, which is denoted as the hazard function $h(t)$ in survival analysis [20]

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t | T > t)}{\Delta t} = \frac{f(t)}{S(t)} \tag{5}$$

The hazard function estimation involves selecting a parametric model, such as the exponential or Weibull distribution, along with an empirical approach. Various hazard shapes, including increasing, decreasing, bathtub, and constant patterns, are commonly observed in practical applications. Fig. 2 illustrates the bathtub curve, often employed to depict product life cycles, comprising three stages [29]: the early failure period, primarily associated with manufacturing defects and transit damage, exhibiting rapid decline; a subsequent random failure period characterized by a stable failure rate; and the wearout period, marked by a sharp increase in the failure rate as products age. Contrary to the assumption of a constant gas pipeline failure rate as discussed in the introduction, which may lack accuracy, this study examines gas pipeline failure rates throughout their life cycles using survival analysis.

2.2.2. Cox model

In contrast to parametric and nonparametric estimation methods, Cox (1972) developed a semiparametric model that assumes the covariates are multiplicatively related to the hazard, and the baseline hazard function is left undefined and not estimated. The general form of the hazard function for the Cox model is as [30]

$$h(t|\mathbf{Z}) = h_0(t)\exp(\beta_1 Z_1 + \beta_2 Z_2 + \cdots + \beta_n Z_n) \tag{6}$$

where $h_0(t)$ is the baseline hazard function used to represent the aging process, and β_i and Z_i are covariates and effect parameters.

The Cox model offers several advantages over parametric and nonparametric models. First, it allows for the estimation of parameters without the need to estimate the baseline hazard function $h_0(t)$, which can be challenging in many cases. Furthermore, the Cox model does not require any specific assumptions about the shape of $h_0(t)$. Instead, it assumes that the covariates are multiplicatively related to the hazard function. The effect parameters can then be estimated using partial likelihood, which involves a product over the observed failure times of conditional probabilities, without the need to involve $h_0(t)$. It also accommodates right-censored data and the likelihood contributions can be appropriately considered, as described in [20].

The survival or failure probability of gas pipelines is of interest in this study, and therefore the baseline hazard function needs to be specified or estimated, allowing it to follow a certain parametric distribution or estimating it nonparametrically. The nonparametric Breslow estimator is adopted here to estimate the baseline survival function $S_0(t)$ [30]. Subsequently, the survival function for an individual pipeline with covariates $\boldsymbol{Z}$ can be obtained

$$S(t) = S_0(t)^{\exp(\boldsymbol{\beta}\mathbf{Z})} \tag{7}$$

The Cox model is a valuable tool for incorporating the effects of various covariates on the failure probability of gas pipelines, and it can effectively address the censoring present in the dataset. However, it is a linear model where the contribution of covariates to the hazard function is modeled using a linear relationship. Therefore, its ability to deal with complex non-linearities, which are often present in the failure mechanisms of gas pipelines, is somewhat limited.

2.2.3. Random survival forests

RSF is an extension of RF designed to handle the right censoring survival data. It consists of random survival trees built on different bootstrap samples. At each node, a random subject of variables is selected, and the node is split based on a survival criterion, the log-rank test as the most widespread one, including survival time and censoring information. For each tree, the cumulative hazard function (CHF) for a terminal node h is calculated using the Nelson–Aalen estimator,

$$H_h(t|\mathbf{Z}_i) = \sum_{t_{l,h} < t} \frac{d_{l,h}}{Y_{l,h}} \tag{8}$$

where $d_{l,h}$ and $Y_{l,h}$ are the number of deaths and individuals at risk at time $t_{l,h}$, and the ensemble cumulative hazard $H_e^*(t|\mathbf{Z}_i)$ is obtained by averaging over all trees in the forest [31]

$$H_e^*(t|\mathbf{Z}_i) = \frac{1}{M} \sum_{m=1}^{M} H_m^*(t|\mathbf{Z}_i) \tag{9}$$

where $H_m^*(t|\mathbf{Z}_i)$ is the CHF for an individual tree, and M is the number of trees.

Harrell's concordance index (C-index) is introduced to estimate prediction errors, which does not depend on a single fixed time for evaluation and can be used to account for censoring. It is defined as the ratio of correctly ordered pairs to comparable pairs. In this context, two samples i and j are considered comparable if the sample with the lower observed time y experienced an event (i.e., $y_i > y_j$ and $s_i = 1$). A pair (i, j) is termed concordant if the estimated risk f by a survival model is higher for subjects with lower survival time (i.e., $f_i > f_j$ and $y_i > y_j$), while the pair is labeled discordant otherwise, as given by [32]

$$C_{index} = \frac{\sum_{i \neq j} I\{\eta_i < \eta_j\} I\{T_i > T_j\} \delta_j}{\sum_{i \neq j} I\{T_i > T_j\} \delta_j} \tag{10}$$

where $I(.)$ denotes the indicator function.

A score value approximating 1 signifies strong discriminatory power between two subjects in terms of event occurrence sequence. RSF is established as a nearly unbiased estimator capable of unveiling intricate variable interrelationships. Additionally, it offers a nonparametric survival function estimate, facilitating the evaluation of covariate effects on survival times. This study employs RSF to model gas pipeline failure probabilities, utilizing both actual failure data and generated data for model training.

2.2.4. Several other machine learning models

1) ANN

ANN simulates biological neural networks and consists of interconnected artificial neurons that transmit signals. The network comprises input, hidden, and output layers, with neurons connected by weighted connections. Neurons activate when the accumulated signal surpasses a threshold, generating output through a non-linear function. Signals propagate through multiple layers for complex computations and feature extraction [33],

$$y = f\left(\sum \boldsymbol{XW} + b\right) \tag{11}$$

where $\boldsymbol{X}$ is the input vector, $\boldsymbol{W}$ is the weight vector, b is the bias term and $f()$ is the activation function.

2) SVR

SVR is a regression technique based on Support Vector Machines (SVM). SVMs are renowned for classification and regression, introduced by Vapnik et al. in 1992 [34]. SVR aims to find a hyperplane function that estimates the connection between predictors and continuous responses. The objective is to minimize errors within a defined margin,

$$y = \sum_{i=1}^{n} \left(\alpha - \alpha_i^*\right) \cdot K(\boldsymbol{x}_i, \boldsymbol{x}) + b \tag{12}$$

where α and α_i^* represent the Lagrange multipliers, while x refers to the input data point for which the prediction is being made, and K denotes the kernel function employed in the analysis.

3) RF

The widely used RF algorithm by Breiman and Cutler [35] combines multiple decision trees to perform classification and regression. RF incorporates feature randomness and selective feature utilization to create uncorrelated trees, reducing overfitting and bias and improving prediction accuracy. RF surpasses individual decision trees and acknowledges the potential advantage of gradient-boosted trees in specific situations. With its adaptability and ability to handle diverse data types, RF is an essential machine learning tool for classification and regression tasks, appreciated for its user-friendliness and ensemble approach.

4) XGBoost

XGBoost, proposed by Chen and Guestrin, is a gradient boosting machine designed for supervised learning problems. It aims to find optimal parameters for mapping training data to their targets. In each iteration, the algorithm builds on previous knowledge by adding a new tree. XGBoost's loss function employs Taylor expansion, optimizing the objective better than first-order approximation. It allows custom loss functions and includes regularization to improve model performance and avoid overfitting [36]. Split finding can be done precisely or roughly. XGBoost is popular for its flexibility, efficiency, accuracy, and scalability.

2.3. Model evaluation

The Cox and RSF models generate survival probabilities ranging from 0 to 1. These results are evaluated by categorizing them into failure or survival outcomes, using a 0.5 threshold in this study. If the predicted survival probability is greater than 0.5, or the failure probability is less than 0.5, the pipeline is considered to be still surviving; otherwise, it is deemed failed. It is important to note that this evaluation is conducted solely on observed failure records.

An additional valuable estimation for the failure analysis of gas pipelines is the mean time to failure, which can be obtained from the survival function as given by

$$T_{mean} = \int_0^\infty t f(t) \mathrm{d}t \tag{13}$$

By leveraging the relationship between the failure probability density function and the survival function and applying integration by parts, Eq. (13) can be further written as

$$T_{mean} = \int_0^\infty S(t) \mathrm{d}t \tag{14}$$

The predicted mean time to failure can be compared with the actual values for each failure record. Several metrics, including mean absolute error (MAE), mean absolute percentage error (MAPE), root mean square error (RMSE), normalized root mean square error (NRMSE), coefficient of determination (R^2) and the Pearson correlation coefficient (ρ) [37] will be calculated to evaluate the performance of the developed failure models. The mathematical expressions for these metrics are as follows

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} \left| T_i - T_{mean,i} \right| \tag{15}$$

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{N} \frac{\left| T_i - T_{mean,i} \right|}{T_i} \tag{16}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \left(T_i - T_{mean,i} \right)^2} \tag{17}$$

$$\mathrm{NRMSE} = \frac{RMSE}{\overline{T}} \tag{18}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} \left(T_i - T_{mean,i} \right)^2}{\sum_{i=1}^{N} (T_i - \overline{T})^2} \tag{19}$$

$$\rho = \frac{Cov\left(T_i, T_{mean,i}\right)}{\sigma_i \sigma_{mean,i}} \tag{20}$$

where Cov is the covariate, σ is the standard deviation, $\overline{T}$ is the mean of the observed failure time and N is the number of data.

For the censored surviving data, a reasonable expectation is the expected remaining life given that an individual pipeline is still surviving at time t_0. In this case, the failure probability at age $t_0 + t$ is as follows

$$P(T \le t_0 + t | T > t_0) = \frac{F(t_0 + t) - F(t_0)}{S(t_0)} \tag{21}$$

The probability density function can thus be derived as

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{F(t_0 + t) - F(t_0)}{S(t_0)} = \frac{f(t_0 + t)}{S(t_0)} \tag{22}$$

Therefore, the expected remaining life can be estimated

$$T_{remain} = \int_0^\infty t \frac{f(t_0 + t)}{S(t_0)} \mathrm{d}t = \frac{1}{S(t_0)} \int_{t_0}^\infty S(t) \mathrm{d}t \tag{23}$$

Given that the actual failure times of censored survival pipelines have not been observed, this study interprets the expected remaining life as a prediction of future failure rates for gas pipelines. It assumes no inspection, maintenance, or repair work will be conducted.

3. Data collection and processing

Reported incidents of gas transmission pipeline failures in the U.S. yield valuable data for predicting future failures. This study analyzes incidents from 1970 to the present, but it is important to note that reporting criteria and forms have changed over time, leading to variations in provided attributes in incident files. Thus, a comprehensive examination and review of failure records is needed to clean and extract relevant information from the data.

3.1. Actual failure records

Each record contains a wealth of information, although not all of it is directly relevant to failures. Data concerning operators, preparers, authorized personnel, and detailed incident consequences (such as total cost, injuries, and fatalities) are omitted. Rather, pertinent attributes are selected and extracted from failure records, using domain knowledge and relevant studies. These attributes encompass the installation year, incident year, pipe material, diameter (D), thickness (t), strength (σ), pressure (P), and location class (LC). The age of failed pipelines is calculated by subtracting the installation year from the incident year,

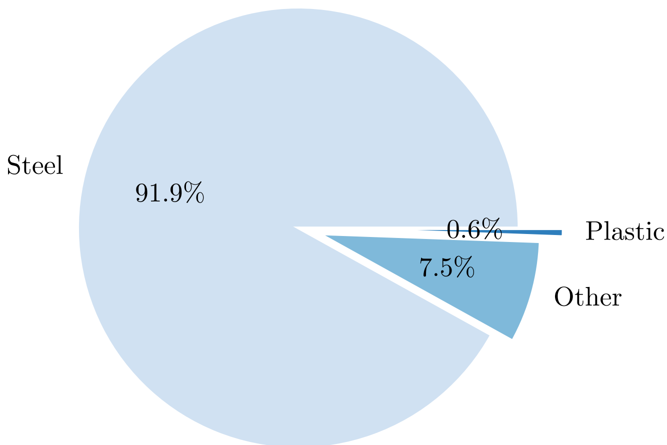


Fig. 3. Distribution of pipeline materials.

which serves as a significant parameter to be modeled in subsequent analyses. Location class refers to the classification of different areas and associated risks based on the potential consequences of pipeline failure, as defined in 49 CFR Section 192.5 [38]. It is determined by quantifying the number of buildings within a specified distance of the pipeline for a mile, thus serving as an environmental factor that may impact gas pipeline operations.

The extracted data underwent a thorough examination to identify outliers likely caused by human error. These records were carefully reviewed and either corrected or eliminated based on evidence from relevant standards and documents. Fig. 3 illustrates the distribution of failed gas transmission pipeline materials, indicating steel as the dominant material, accounting for over 91.9% of gas pipelines. Consequently, this study focuses solely on developing failure models for steel gas transmission pipelines, excluding records involving other materials. It is important to note that a single pipeline may experience incidents, undergo repairs, and resume service, potentially resulting in multiple failures over its operational lifespan. However, this study solely analyzes the initial failure event, disregarding subsequent failures. The resulting dataset consists of 6386 failure records. Additionally, some records have missing data, and excluding them during modeling could lead to valuable information loss. To address this, the MissForest algorithm, known for its superior performance, was utilized to impute the missing data.

3.2. Data generation for pipelines in service

Including right-censored survival pipelines is crucial in developing unbiased survival models. However, due to confidentiality, PHMSA does not provide data for operational gas transmission pipelines. Fortunately, annual report data from PHMSA can be utilized to generate parameters for right-censored survival pipelines. These reports offer information on the total length of gas pipelines and their breakdown by diameter, material, installation decade, and location class. This data allows estimation of the number of installed pipelines, pipe diameter, and location class for each year. By considering this information, parameters for right-censored survival pipelines can be roughly generated to enhance model performance.

Pipe thickness and strength have been shown to be correlated with pipe diameter in previous studies [39] and in this study. It was assumed that the distribution of pipe diameter and strength followed a normal distribution. The values for pipe diameter and strength were generated based on their respective distributions, using mean and variances estimated from the correlation with each specified pipe diameter. However, it is important to note that pipe thickness and strength were not generated arbitrarily. Standards such as API specification 5L and ASME B31.8 provide guidelines for allowable pipe wall thickness and yield strength based on pipe diameter. Therefore, in this study, the range of pipe thickness and strength was limited accordingly, and a truncated normal distribution was used. The expression for this distribution is given by

$$f(x;\mu,\sigma,a,b) = \frac{1}{\sigma}\frac{\phi\left(\frac{x-\mu}{\sigma}\right)}{\phi\left(\frac{b-\mu}{\sigma}\right)-\phi\left(\frac{a-\mu}{\sigma}\right)} \tag{24}$$

where a and b are upper and lower bounds, μ and σ^2 are mean and

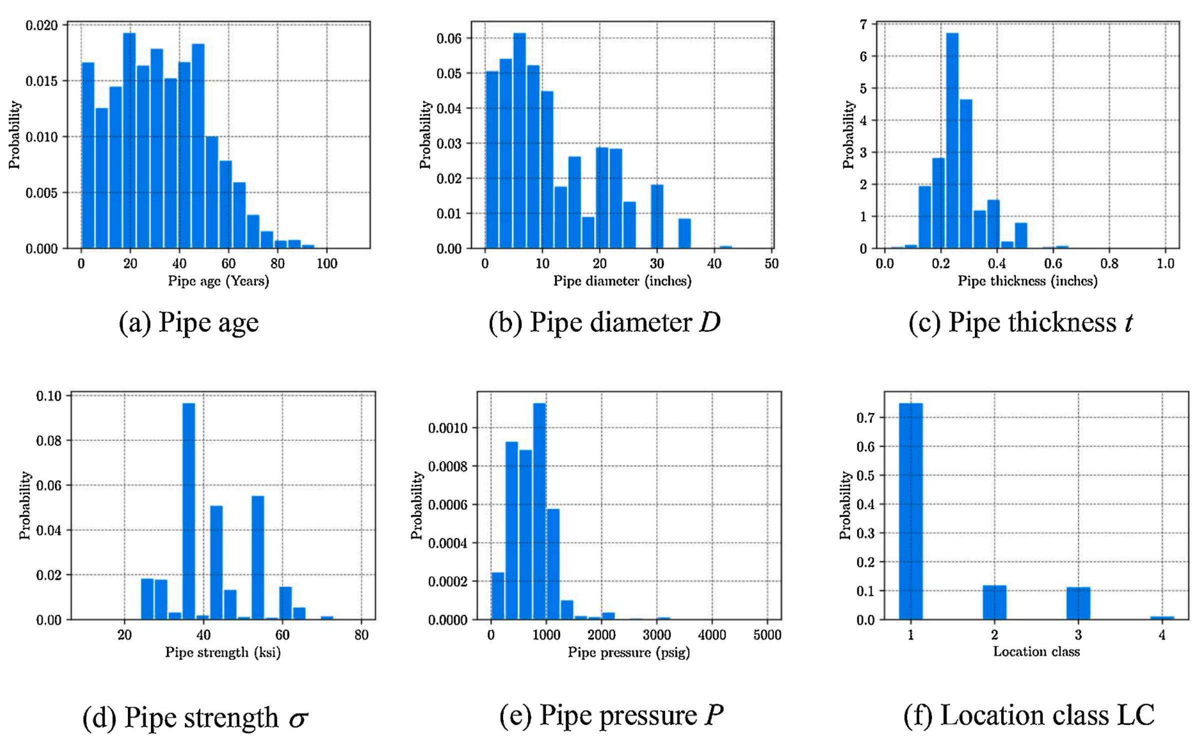


Fig. 4. Statistical distributions of factors of failure records.

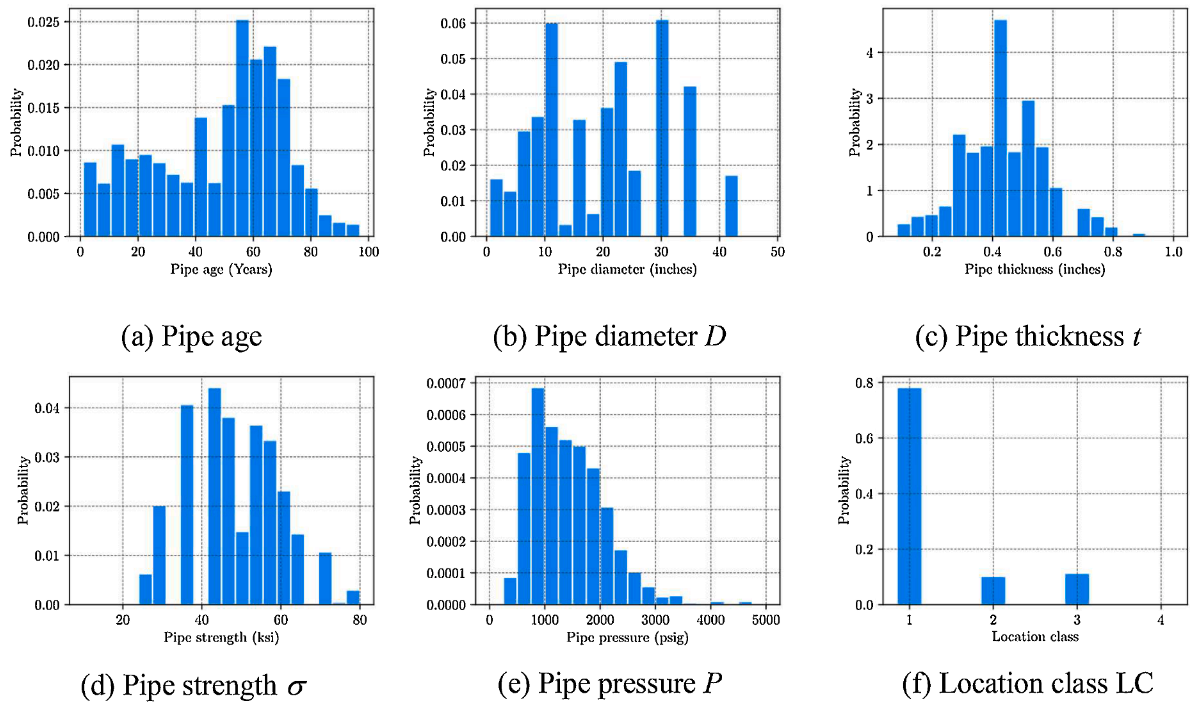


Fig. 5. Statistical distributions of factors of generated pipelines.

variance respectively, ϕ is the probability density function of the standard normal distribution.

To ensure the generated data closely reflects actual conditions, the evolution of pipeline characteristics is considered. Historical developments, such as the discontinuation of certain grades and the addition of new pipelines, are taken into account. For example, Grade C with a yield strength of 45 ksi was discontinued in the 1930s, and Grade X46 and X52 pipelines were added in 1953. Furthermore, the introduction of large diameter pipelines up to 30 inches occurred in 1946. These historical factors contribute to the practicality of the generated survival pipelines. Comparisons between the distribution of attributes for the generated pipelines, such as installation year, pipe diameter, and location class, and the corresponding statistics from annual records demonstrate a notable similarity. This provides evidence supporting the accuracy of the generated attributes for in-service gas pipelines. The pressure of the pipelines is determined using the design formula specified in CFR 192.105 [38],

$$P = \frac{2St}{D} EFT \tag{25}$$

where D is the nominal outside diameter in inches, t is the nominal wall thickness, S is yield strength, E is the longitudinal joint factor, F is the design factor and T is the temperature derating factor. The exact values of these factors can be found in the reference [38].

3.3. Data comparison

The statistical distributions of failure record factors before data imputation and generated pipelines are presented in Figs. 4 and 5, respectively. Notably, significant differences in the distribution of pipe age, diameter, thickness, and pressure are apparent between failed and operational pipelines. Fig. 4(a) demonstrates that a proportion of gas pipelines experienced failures in their early stages, aligning with the initial failure period outlined by the bathtub curve. Furthermore, a

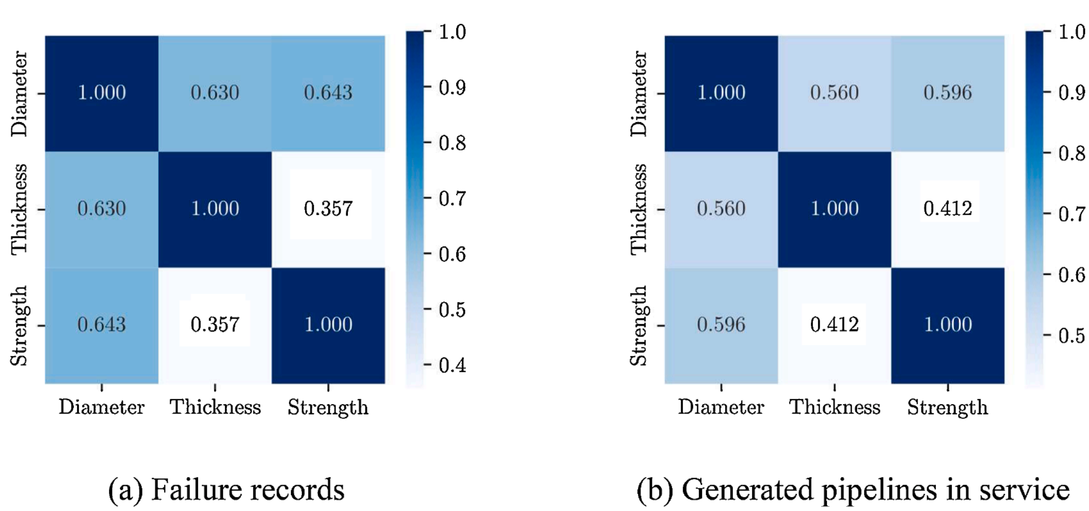


Fig. 6. Correlation coefficients between factors.

Table 1
Data description.

	Material	Period	Number
Failure records	Steel	1970 to the present	6386
Censored pipelines		From installation to the present	24564

Table 2
Inputs and outputs of developed models.

Model	Input	Output
Cox, RSF	D, t, t / D, σ, P, LC	Survival probability
ANN, SVR, RF, XGBoost	D, t, t / D, σ, P, LC	Failure age

majority of the failed pipelines possessed an age distribution below 60 years. Conversely, the pipelines still in service were primarily installed in the 1950s and 1960s, suggesting a long operational lifespan of approximately 50 to 70 years. Figs 4**(b)** and 5**(b)** reveal that smaller diameter pipelines are more vulnerable to damage compared to their larger diameter counterparts, which are predominantly operational. Notably, failed pipelines generally exhibit smaller thickness, centered around 0.2 inches, while in-service pipelines exhibit thicker walls, centered around 0.4 inches. This finding suggests that pipeline thickness is a crucial factor to consider when analyzing and modeling gas pipeline failures. Moreover, operational pipelines tend to operate at higher pressure levels, attributed to their larger diameter and thicker walls, enabling them to have a higher natural gas delivery capacity. It is essential to emphasize that both failed and operational pipelines are predominantly laid in the 1st location class. Additionally, it is noteworthy that larger diameter pipelines were installed more recently, resulting in a relatively shorter duration of service.

Fig. 6 depicts the correlation matrix computed using Eq. (20) for the two datasets. As observed from the failure records, a good correlation relationship still exists between the pipe characteristics in the generated dataset. In the following section, generated dataset and failure records were utilized to develop failure probability models. Table 1 summarizes the data used in the subsequent analysis.

4. Model development and results

4.1. Survival and machine learning models

The study utilized a prepared dataset to develop and validate various models, including the Cox model, RSF model, and ANN, SVR, RF, and XGBoost models. The first two models used the complete dataset for training and validation, aiming to predict the probability of survival or failure for each pipeline. Additionally, Eq. (14) was employed to estimate the mean time to failure for each failure record based on these predictions. On the other hand, the ANN, SVR, RF, and XGBoost models were trained using failure records solely since they cannot handle right-censored data directly. These models were used to predict the estimated failure time for each pipeline, which could be compared with the estimated mean time to failure generated by the Cox and RSF models. The influence of gas pipeline relative thickness on failure patterns has been demonstrated [40], therefore, prompting the computation of the thickness-to-diameter ratio and its incorporation into the feature vector to enhance the performance of the developed models. Consequently, a total of six features were utilized: pipe diameter, thickness, thickness-to-diameter ratio, strength, pressure, and location class.

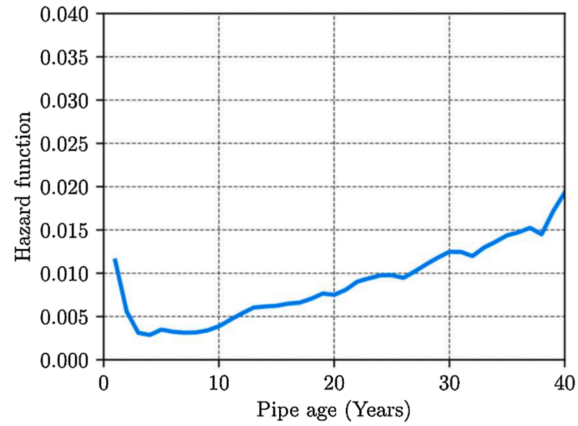


Fig. 7. The estimated baseline hazard function.

Table 2 provides an overview of the inputs and outputs of the developed models. To ensure consistency, failure records before 2010 and the generated pipelines observed up to 2010 served as the training and development data. Validation purposes solely employed failure records occurring after 2010, while censored survival pipelines were used as a reference to predict the failure rate for the following ten years.

To enhance the performance of machine learning models, the grid search method is employed to systematically explore various combinations of hyperparameters. This method is commonly utilized to identify the best hyperparameters that yield optimal performance for a given model. For the RSF model, the hyperparameters are as follows: the number of estimators is 200, the minimum number of samples required to split a node is 3, and the minimum number of samples required to be at a leaf node is 1. The bootstrap is utilized to select samples for building each tree. In the ANN model, a three-hidden layer structure is adopted, containing 40 neurons in each layer. The rectified linear unit (ReLU) is chosen as the activation function. In the SVR model, the radial basis function (RBF) is used as the kernel function with a regularization parameter (C) set to 60 and a kernel coefficient of 0.1. The RF model generates 200 decision trees, each with a maximum depth of 12. The criteria for splitting nodes include a minimum of 2 samples, and each leaf node must have at least 1 sample. Moreover, the number of features randomly selected is typically the square root of the total number of features. For the XGBoost model, 200 decision trees are employed, with a maximum depth of 6 for each tree. The learning rate is set to 0.05, and the minimum sum of sample weights for child nodes is 1.

4.2. Results

4.2.1. Cox model

Table 3 shows the estimated parameters of each covariate in Eq. (6), which help in understanding their impact on the likelihood of gas pipeline failures. The covariates influence the baseline hazard by a factor of $\exp(\beta)$, with a larger positive coefficient indicating a higher risk to the gas pipeline. Notably, pipelines with a larger diameter, thicker thickness, and higher yield strength are associated with a reduced risk of failure. Conversely, pipe pressure and location class increase the risk of failures, which aligns with expectations. For instance, a gas pipeline placed in a higher location class may encounter greater human, traffic, and environmental disturbances, hastening its deterioration process. Fig. 7 illustrates the estimated baseline hazard function for the model

Table 3
Covariates and estimated coefficients in the Cox model

Covariate	Diameter	Thickness	Thickness / Diameter	Yield strength	Pressure	Location class
Coefficient β	-5.076×10^{-2}	-5.718	-0.530	-5.750×10^{-5}	3.999×10^{-4}	1.896×10^{-2}

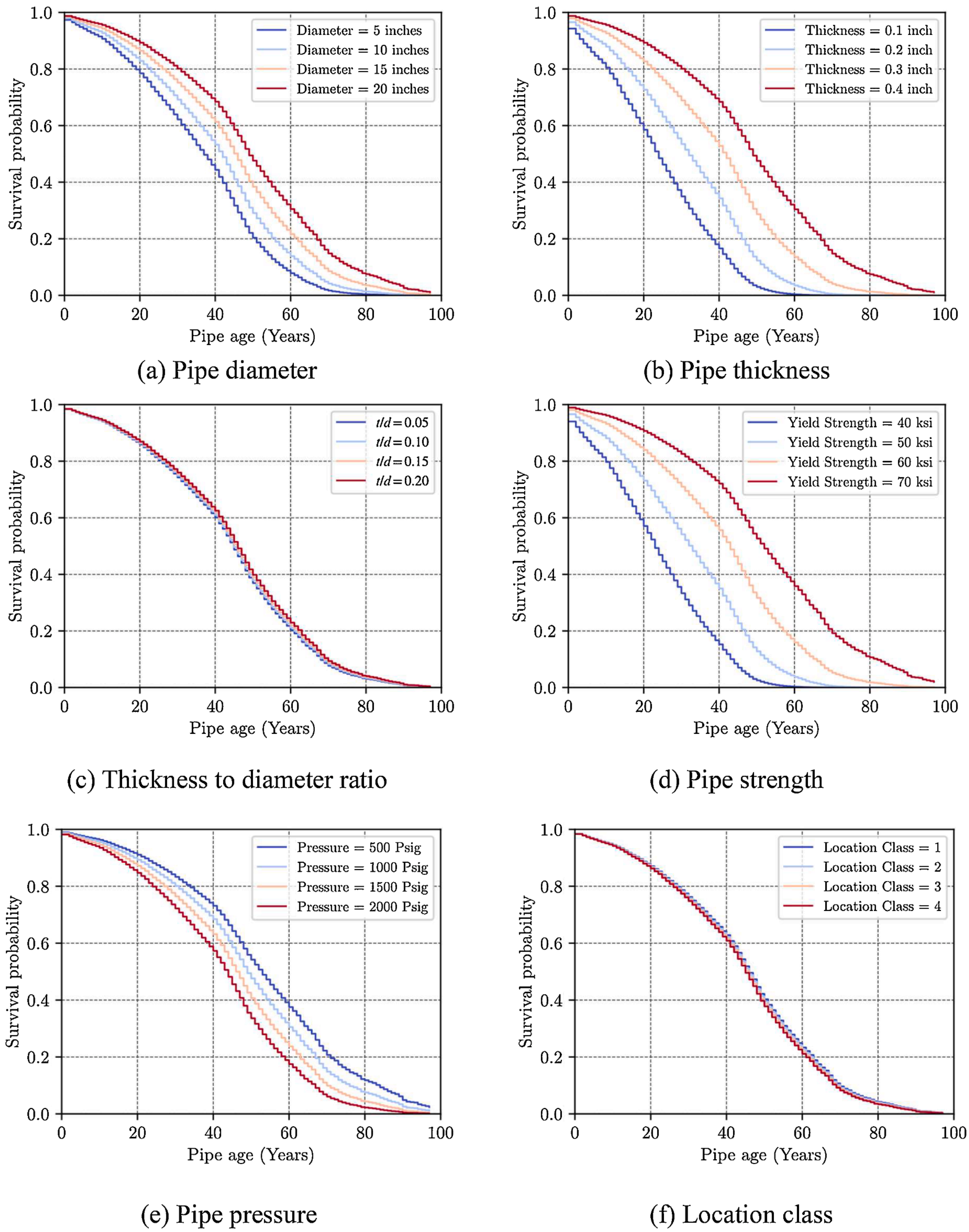


Fig. 8. Survival probability of different covariates predicted by the Cox model.

described in Eq. (6). The hazard function displays a heightened risk of failure in the initial years, which rapidly declines. However, the hazard rate gradually rises with pipe age, resembling the typical bathtub curve pattern. This finding suggests that assuming a constant failure rate when modeling gas pipeline failures may not be suitable. Instead, employing the bathtub curve pattern in hazard modeling would better capture the changing risk of failure over time.

The impacts of covariates on the failure of gas pipelines can be better understood visually through the failure probability or survival probability. In this study, the survival probability is chosen for analysis, and the failure probability can simply be calculated by subtracting the survival probability from 1 as Eq. (3). The Breslow estimator is used to estimate the nonparametric baseline survival function $S_0(t)$, which is then used in Eq. (7) to estimate the survival probability. Fig. 8 illustrates the survival probability of pipelines with varying covariates over time. Consistent with previous findings, pipelines with smaller diameters, thinner walls, and lower yield strength exhibit higher probabilities of failure within a few decades. It is important to note that physical

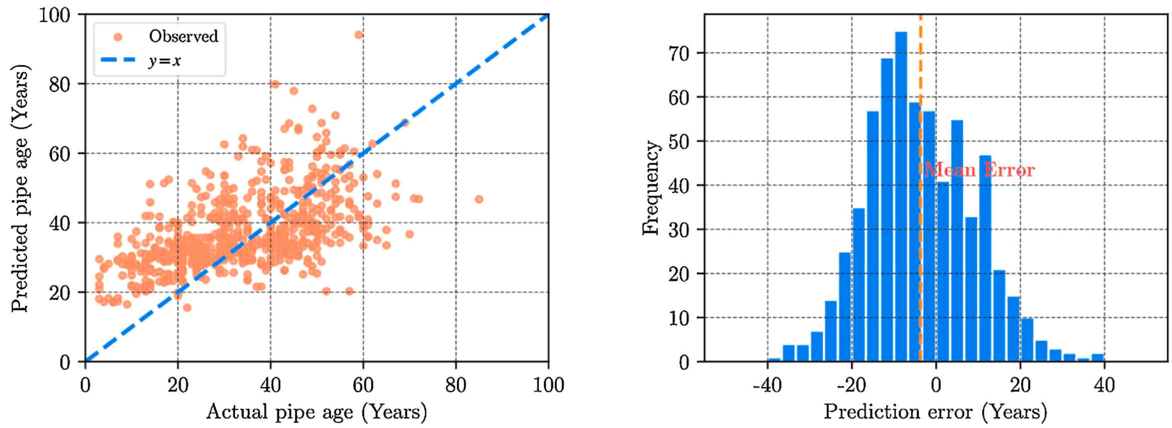


Fig. 9. (a) Scatter plot, and (b) error (observed − predicted) histogram, of the predicted mean time to failure using the Cox model and actual pipe age.

covariates such as diameter, thickness, and strength have the most substantial impact on survival probabilities. The influence of the thickness-to-diameter ratio and location class on survival probability is also observed, but the effects are less pronounced compared to the aforementioned covariates.

The mean time to failure for individual pipelines can be estimated using the fitted Cox model, as indicated in Eq. (14). Fig. 9 presents a scatter plot illustrating the predicted mean time to failure and the actual pipe age for the validation data. Additionally, an error histogram is provided. It is worth noting that the predicted mean time to failure for each pipeline does not align well with the actual data. In fact, the Cox model tends to overestimate the failure age of gas pipelines. This is evident in Fig. 9**b**, which highlights a maximum absolute error of 39 years. One reason for this overestimation is the inclusion of right-censored pipelines during model development. It is observed, in Fig. 4**a**, that a certain percentage of these pipelines have been in service longer than the age of many failed pipelines. Moreover, the limitation of the Cox model as an essentially linear model further contributes to this issue, as it fails to capture the complex nonlinearity associated with the deterioration of gas pipelines. Nonetheless, the Cox model does allow for a preliminary analysis of the impact of covariates on the survival function or the hazard function. Additionally, it has the capability to derive an analytic expression for the survival probability by employing a parametric form to fit the baseline hazard when necessary.

4.2.2. RSF model

The RSF model is employed to estimate the survival and cumulative hazard functions non-parametrically for individual pipelines. The prediction obtained by averaging across all trees is evaluated on the validation data, resulting in a C-index of 0.818, which indicates a promising performance. This suggests that the model has good discriminatory power in distinguishing between pipelines based on survival times. The detailed insight provided by Fig. 10 illustrates the predicted survival probabilities over pipe age with different covariates. The impacts of these covariates on the survival probability align with the results obtained from the Cox model. However, unlike the Cox model, the RSF model presents these impacts implicitly in a more complex nonlinear manner, lacking a simple analytic expression. Feature importance is measured using permutation, and the results in Fig. 11 highlight the relative importance of factors influencing gas pipeline deterioration. Thickness emerges as the most critical factor, followed by strength, pressure, and the thickness-to-diameter ratio. These findings underscore the crucial roles of these factors in pipeline degradation.

The mean time to failure for individual pipelines was calculated using Eq. (14), and the resulting scatter plot and error histogram are displayed in Fig. 12**(a)** and **(b)** respectively. A comparison between the predicted results in Fig. 12 and those in Fig. 9, obtained through the Cox model, reveals a significant enhancement. In Fig. 12, the predicted results exhibit a more linear relationship with the ideal $y = x$ line. The corresponding R^2 and ρ values, based on Eqs. (19) and (20), are reported as 0.810 and 0.902 respectively. Moreover, the error histogram reveals that the largest absolute error is only around 20 years. This improvement can be attributed to the RSF method's capability to offer a nonlinear approximation of the statistical relationships between factors and pipeline failures. Furthermore, the prediction errors approximate a normal distribution with a mean close to 0, further illustrating the effectiveness of the RSF method in modeling gas pipeline failures or survival probabilities, particularly when dealing with censored data. Overall, these findings underscore the strengths and capabilities of the RSF method in modeling gas pipeline failures.

In the validation dataset, the status of pipelines is determined based on Section 2.3, and the survival probability is estimated using the Cox and RSF models. Subsequently, the target survival probability or failure probability is calculated considering the number of years each pipeline has survived and its current status. The accuracy of the prediction results over the study period and for the next decade is presented in Table 4. The results indicate that approximately 61.68% and 80.53% of the pipelines are estimated to fail over the study period by the Cox and RSF models, respectively. Moreover, a higher percentage of 88.16% and 94.86% is predicted to fail in the next decade. The RSF model outperforms the Cox model in both predictions, emphasizing the superiority of the RSF method in modeling gas pipeline failure.

4.2.3. Comparison with other machine learning models

Multiple machine learning models were developed to accurately predict the failure age of individual pipelines using only failure records. To maintain consistency, the same subset of failure records used to train the RSF model was also employed for training these additional models, while the remaining records were reserved for validation purposes. All four machine learning models underwent hyperparameter optimization using the grid search method. The results obtained from these models were utilized as a benchmark to evaluate the performance of the other two models. Fig. 13 depicts the predicted ages of the pipelines alongside their actual ages, showcasing a significant improvement compared to the results obtained from the Cox model in Fig. 9. Furthermore, a nearly linear relationship was observed in the scatter plot, with respective Pearson correlation coefficients of 0.702, 0.722, 0.821, and 0.828. Additionally, the maximum absolute error was reduced when compared to the Cox model. Among the four machine learning models assessed in this study (RF, XGBoost, ANN, and SVR), the ensemble methods of RF and XGBoost consistently outperformed ANN and SVR.

However, the performance of the four machine learning models is slightly inferior to that of the RSF model. The prediction errors are distributed more widely around 0, unlike the more concentrated errors

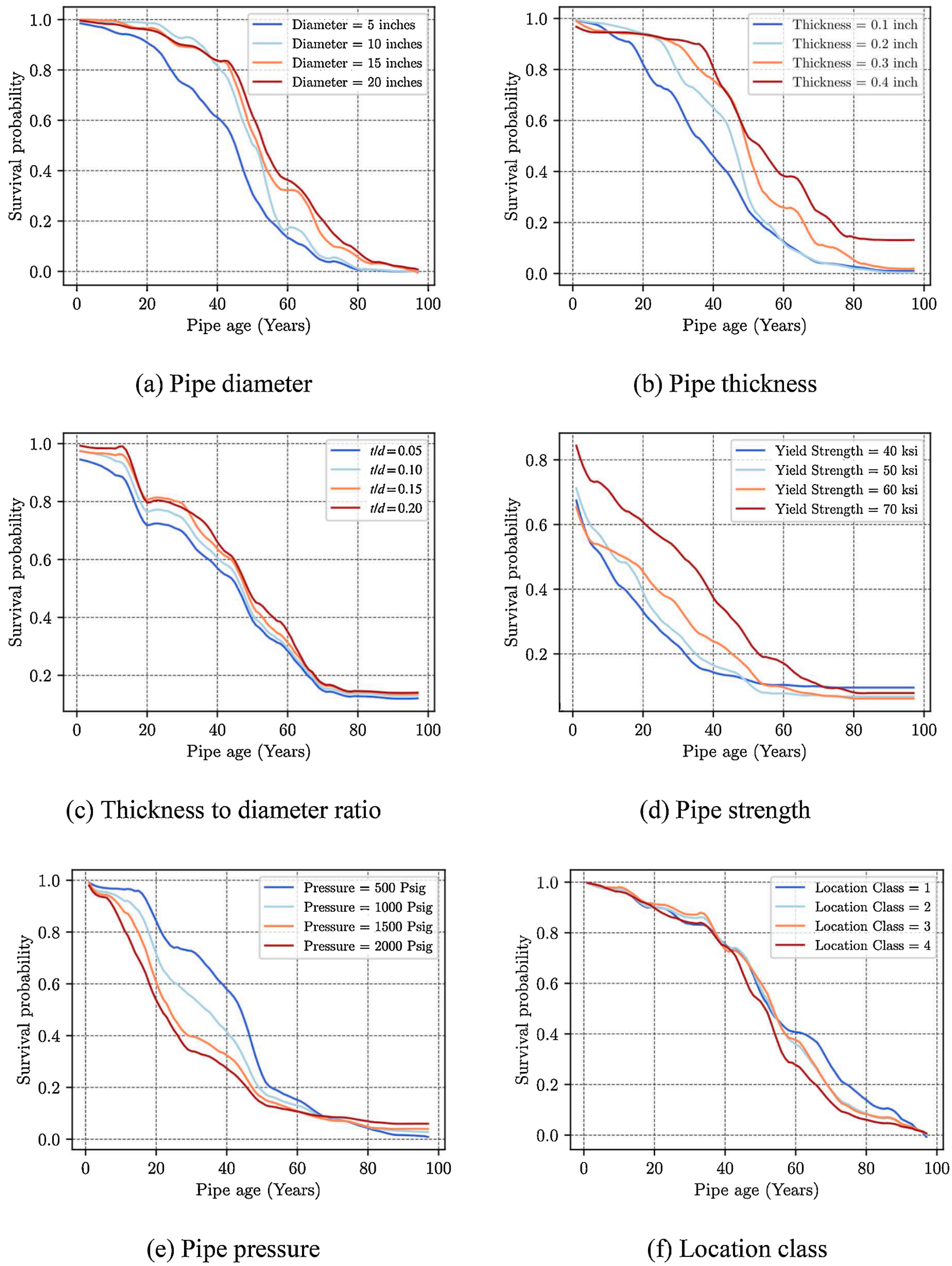


Fig. 10. Survival probability of different covariates predicted by RSF model.

observed in the results of the RSF model. The mean prediction errors for the four machine learning models are 1.480, 0.649, 0.371, and 0.570, respectively, deviating from the ideal mean, which also suggests that the machine learning model generally underestimates the failure age of gas pipelines. This is because the traditional machine learning models are not suitable for directly incorporating the information of censored pipelines into their model development.

This section provides an evaluation of the developed models by comparing the mean time to failure estimated by various models including Cox, RSF, ANN, SVR, RF, and XGBoost. Table 5 presents evaluation metrics such as MAE, MAPE, RMSE, NRMSE, R^2 and ρ based on Eqs. (15) ~ (20). The RSF model exhibits superior performance

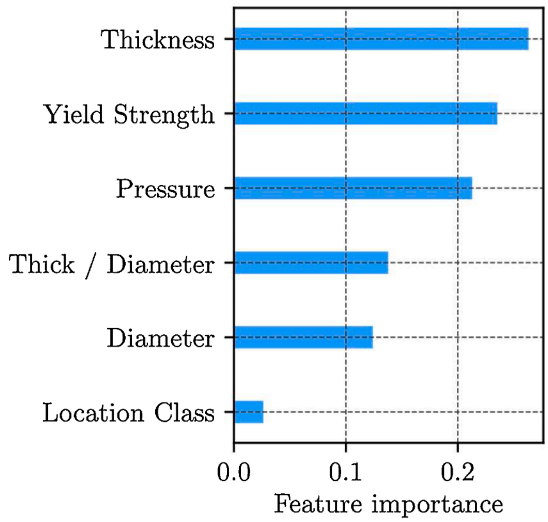


Fig. 11. Feature importance.

compared to other models, with the lowest values for MAE (4.835), MAPE (0.199), RMSE (6.438), and NRMSE (0.192). Additionally, it achieves the highest R^2 (0.810) and ρ (0.902) values. Conversely, the Cox model performs the worst, while all other machine learning models outperform it. Among commonly used machine learning models, ensemble methods such as RF and XGBoost show similar performance, surpassing ANN and SVR. This suggests that machine learning models are more suitable for modeling gas pipeline failures based on historical records, as they can capture the complex nonlinearity between factors and failure. However, incorporating censored data, as done in the RSF model, further improves performance. Furthermore, the RSF model enables the prediction of the survival probability of individual pipelines over time, providing a more comprehensive understanding of the deterioration process, rather than solely focusing on failure age.

This study further evaluates the prediction results of developed models by applying them to a representative set of pipelines. Table 6 provides a comprehensive overview of the characteristics of these pipelines, including different diameters and location classes. Subsequently, the developed models are used to predict the service age of these pipelines. The accuracy of the models is evaluated by comparing the observed age of the pipelines to the predicted age, as depicted in Fig. 14. A ratio close to 1 indicates better model performance. The RSF model consistently outperforms the other models in most cases, with ratios close to 1. However, the fourth pipeline shows the poorest prediction performance, although it still outperforms all other models. Interestingly, XGBoost performs exceptionally well for the third and fifth pipelines, slightly surpassing the RSF model, although it occasionally underperforms, similar to the first pipeline. RF also demonstrates relatively strong performance, while ANN and SVR generally exhibit lower levels of accuracy compared to the other models. These findings once again underscore the importance of incorporating censoring information in gas pipeline failure modeling and highlight the superior performance of the RSF model.

4.2.4. Prediction of failure rate for the next decade

This section presents an interesting study that aimed to estimate the remaining lifespan of censored pipelines using the conditional survival probability, as described in Eq. (23). The obtained results were then utilized to estimate the likely failure rate of gas transmission pipelines in the U.S. However, it is important to acknowledge that these estimates were derived from a simulated dataset, thus serving as reference rather than definitive predictions. Fig. 15 displays the predicted failure rates of gas pipelines for the next decade using the Cox and RSF models, assuming no maintenance work. Both models indicate a gradual increase in the failure rate over the next ten years. This outcome aligns with expectations, as the majority of in-service pipelines were installed in the 1950s and 1960s and have been in service for a considerable duration, thereby increasing their vulnerability to failure as they approach the wearout phase. The results of this study underscore the significance of enhancing safety and risk management measures for pipelines to minimize the potential impact on society.

4.3. Discussion

The purpose of this study was to propose a methodology for modeling the failure of gas transmission pipelines in the U.S. However, to adhere to regulatory requirements, failure records prior to 1970 were inaccessible and thus excluded from the dataset. Additionally, changes in incident reporting criteria throughout the years meant that not all incidents were reported to PHMSA, potentially introducing bias to the dataset. Consequently, slight disparities may exist between the estimated hazard function and the actual one due to the missing data and potential bias. Moreover, as data on currently operational gas transmission pipelines is confidential, this study relied on generated data based on annual reports from PHMSA. Although the generated dataset

Table 4
Estimated failure portion of the actual failure records of gas pipelines after the year 2010 by Cox and RSF models.

	Over the study period	In the next decade
Cox	61.68%	88.16%
RSF	80.53%	94.86%

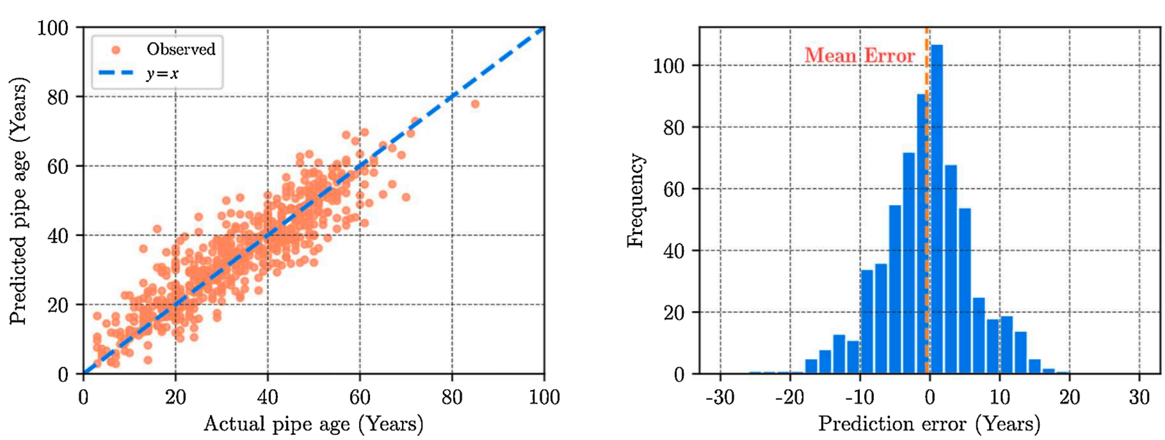


Fig. 12. (a) Scatter plot, and (b) error (observed − predicted) histogram, of the predicted mean time to failure using the RSF model and actual pipe age.

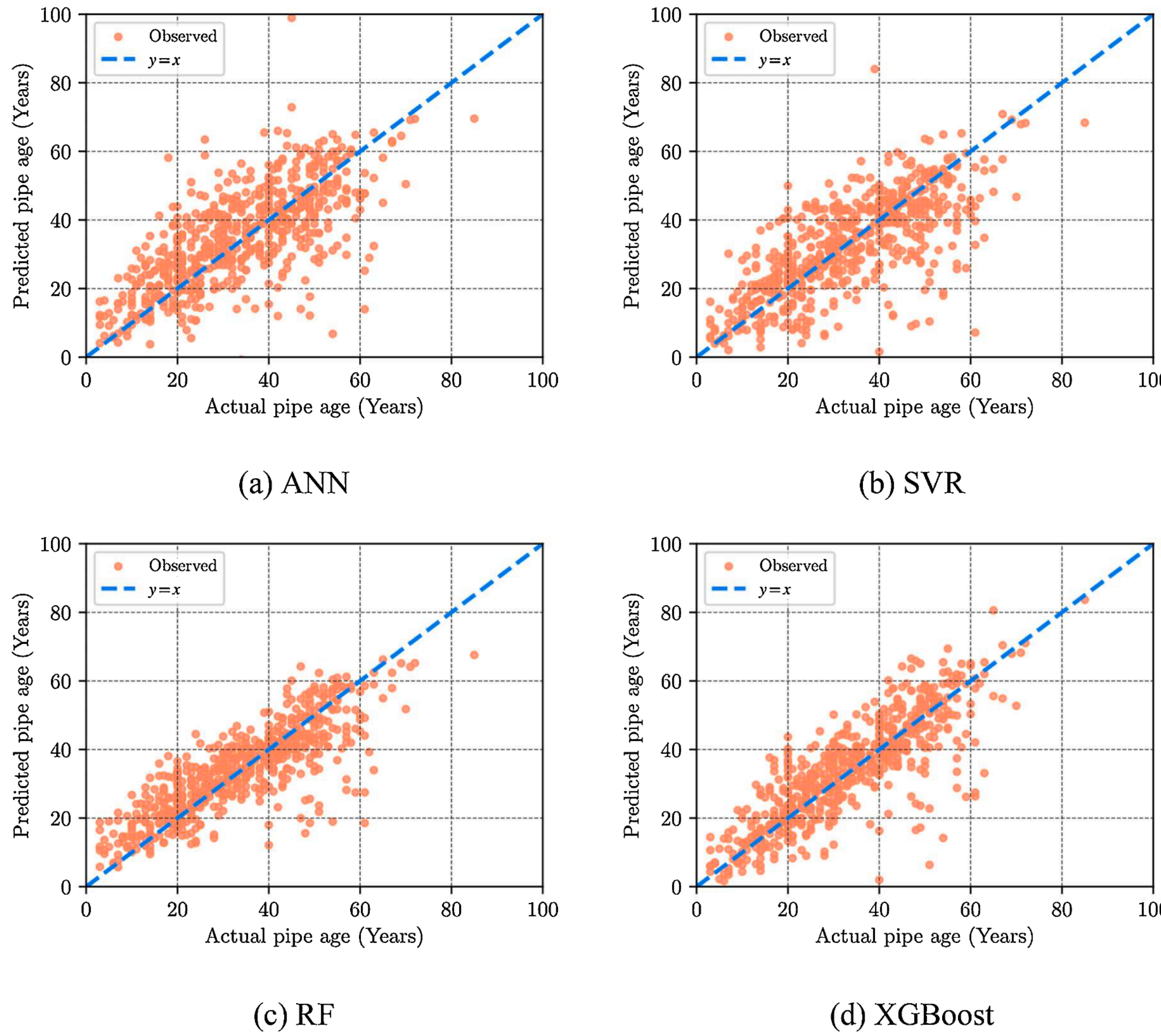


Fig. 13. Scatter plots of predicted mean time to failure by various machine learning models compared to actual pipe age.

Table 5
Performance evaluation of various developed models

	MAE	MAPE	RMSE	NRMSE	R^2	ρ
Cox	10.957	0.529	13.271	0.397	0.191	0.538
RSF	4.835	0.199	6.438	0.192	0.810	0.902
ANN	8.541	0.337	11.306	0.338	0.413	0.702
SVR	7.780	0.293	10.770	0.322	0.467	0.722
RF	6.195	0.260	8.441	0.252	0.673	0.821
XGBoost	6.188	0.235	8.757	0.262	0.648	0.828

aimed to closely resemble the actual data by considering U.S. standards, construction history, and domain knowledge, discrepancies are inevitable, leading to potential errors in the modeling process. It is essential to recognize that the aforementioned limitations emphasize that while the RSF model exhibits stronger performance compared to alternative models for failure modeling, there is room for further enhancement.

The analysis in this study was limited by the availability of data. Several important factors that could have influenced the results, such as pipe length, cathodic protection, pipe coating, buried depth, soil properties, and environmental conditions (temperature, precipitation, humidity, etc.), were not included in the analysis due to missing data on failure records or in-service pipelines. Additionally, valuable information from historical inspection data, including corrosion defect attributes and maintenance records, could have enhanced the dataset and improved the performance of the developed models. Unfortunately, these data sources were not accessible for this study. To improve the accuracy and applicability of gas transmission pipelines, which exhibit geographical diversity and vary in characteristics, refinement of the developed models is necessary. Accessing comprehensive information on in-service pipelines and obtaining more extensive data on failed pipelines would contribute to this refinement. However, acquiring such

Table 6
Attributes of representative gas transmission pipelines.

	Installed Year	Diameter	Thickness	Thickness / Diameter	Strength	Pressure	Location Class	Observed Failure Age
1	1952	8	0.3	3.75×10^{-2}	42000	2000	1	30
2	1959	20	0.25	1.25×10^{-2}	52000	936	1	33
3	1964	36	0.344	9.55×10^{-3}	60000	800	1	47
4	1970	4	0.12	3.00×10^{-2}	35000	760	2	13
5	1954	16	0.314	1.963×10^{-2}	35000	110	4	29

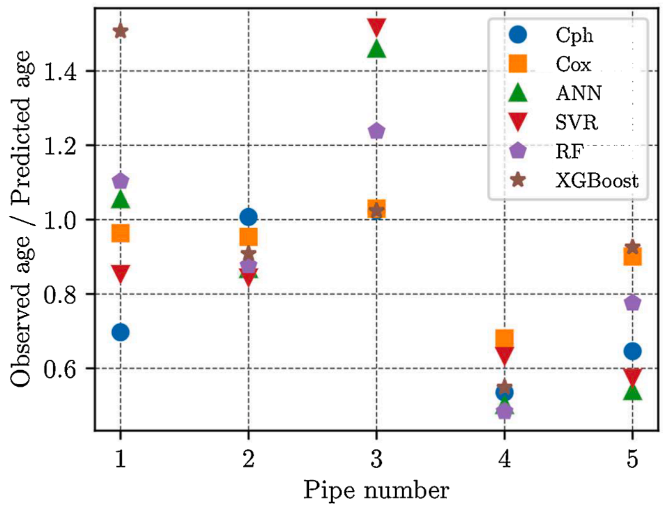


Fig. 14. Comparison of observed age to predicted age ratios for representative pipelines using various models.

data is currently not feasible. Despite this limitation, the models developed using the proposed methodology and existing dataset yielded satisfactory results, comparable to studies conducted on small-scale water pipe networks with more complete data [24,41]. The practical significance of these models lies in their potential to guide integrity management, maintenance activities, and risk analysis of gas transmission pipelines.

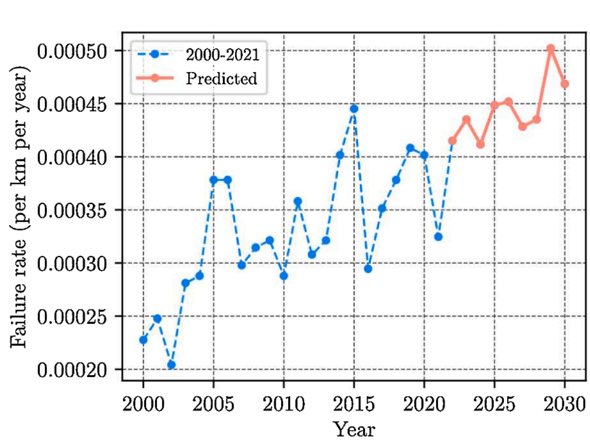
This study primarily focuses on analyzing a database of pipeline failures in the US, but the proposed method can be adaptable to different subjects with varying properties, including gas gathering systems, water pipelines, or structural components. However, it is crucial for the events to share similar characteristics or have consistent definitions across all subjects to apply the proposed approach successfully. This requires the observation or monitoring of subjects to identify the occurrence of an event of interest within a specified timeframe. Accurate and comprehensive data regarding the timing of events for each subject is necessary. Additionally, subjects should be monitored for a sufficient duration to capture the event of interest. The length of the follow-up period should be determined based on the event's characteristics and research objectives, allowing for sufficient time to observe a significant number of events. Meeting these prerequisites enables the proposed method from this study to be promising for modeling the failure process in various applications.

5. Conclusion

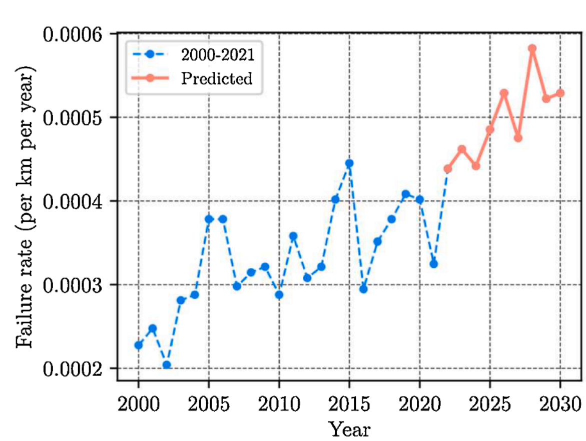
This study presents a methodology to establish an efficient and reliable model for predicting failures in gas transmission pipelines using historical failure data. Statistical, i.e., Cox, survival and machine learning integrated, i.e., RSF, and machine learning models, i.e., ANN, SVR, RF and XGBoost, are developed on the prepared dataset. The Cox and RSF models estimate the survival probability for each pipeline throughout its life cycle, considering the censoring information in the dataset. The ANN, SVR, RF, and XGBoost models use failure data solely to predict the failure age for each pipeline and are compared with the results obtained from the Cox and RSF models.

Cox and RSF models reveal that physical covariates, such as pipe thickness and strength, significantly affect the survival probability of pipelines. Thicker and higher-strength pipes have a greater survival probability. However, pipelines operating at higher pressures and in higher location classes face a higher risk of failure. The hazard function follows a bathtub curve pattern, indicating varying failure risks at different stages of the pipeline's life, underscoring the significance of incorporating the dynamic failure rate in relation to pipeline age when developing models. The Cox model tends to overestimate pipeline life and struggles with nonlinearities. Conversely, other machine learning models tend to underestimate failure ages due to their incapability to handle censored pipelines. In contrast, the RSF model outperforms both the Cox model and other machine learning models because it effectively incorporates the nonlinear characteristics of pipeline failure and censoring information. This highlights the significance of properly addressing censoring in the modeling approach. The proposed methodology in this study accurately models failure patterns using censored information and enables the evaluation of covariates' impact on failure probability and the dynamic failure rate. Consequently, this method improves the accuracy of failure prediction for gas pipelines, providing valuable insights for enhancing safety and risk management practices.

The models' predictive power relies on having extensive records, inspection and maintenance records. Having comprehensive data on pipeline failures and a larger, diverse dataset can further improve the models' performance. However, obtaining such information requires well-established standards, procedures, and a well-maintained database, which can pose challenges and potentially impact the proposed methodology's effectiveness. Nevertheless, the developed model using the proposed methodology has shown promising results. Additionally, the proposed method possesses potential for application in other systems with reliability practices, provided certain prerequisites are met. It is anticipated that future studies will explore its implementation in various contexts.



(a) Cox model



(b) RSF model

Fig. 15. Predicted failure rate of gas transmission pipelines in the U.S.

CRediT authorship contribution statement

Rui Xiao: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft. **Tarek Zayed:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – review & editing. **Mohamed A. Meguid:** Supervision, Investigation, Writing – review & editing. **Laxmi Sushama:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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